

ICMP Flood Denial-of-Service Attack

NIST Cybersecurity Framework — Incident Analysis & Response Plan

Report ID: INC-2024-0047

Date filed: May 27, 2026

Analyst: Cybersecurity Team

Severity: High

Status: Resolved

EXECUTIVE SUMMARY

On May 20, 2026, the organization's internal network experienced a denial-of-service (DoS) attack lasting approximately two hours. A malicious actor exploited an unconfigured firewall by flooding the network with ICMP echo request packets, rendering all internal network services inaccessible to staff. The incident management team contained the attack by blocking incoming ICMP traffic and restoring critical services. A series of remediation controls have since been implemented to prevent recurrence.

INCIDENT TIMELINE

09:00 AM: Unusually high volume of inbound ICMP echo request packets detected targeting the perimeter firewall.

09:15 AM: Automated monitoring alerts trigger due to high network latency; staff report widespread inability to access internal web design and graphic design workflows.

09:25 AM: Incident Management Team is assembled and begins triage.

09:45 AM: Analysts successfully isolate the root cause to a spoofed external ICMP flood bypassing the unconfigured perimeter firewall.

10:15 AM: Emergency firewall rules are implemented to drop all inbound ICMP traffic and apply rate-limiting. Network load begins to stabilize.

11:00 AM: Non-critical services are incrementally restored. Critical services are verified as fully operational.

11:15 AM: Incident declared contained and closed. Post-incident monitoring phase begins.

Govern	NIST CSF —GV
Organizational Context & Risk Strategy Prior to this incident, network configuration changes were made on an ad-hoc basis without formal oversight, leading to the misconfigured firewall. Leadership is now implementing a structured risk management strategy to align IT operations with organizational risk tolerance.	
Governance Improvements & Policy Actions Firewall Change Management Policy: Establish a formal approval workflow requiring	

secondary sign-off by the IT Security Lead before any configuration changes are applied to edge devices.

Executive Oversight: The Cybersecurity Team will present a high-level network security posture report to senior management on a quarterly basis, highlighting detected anomalies and patch compliance.

Role Accountability: A dedicated Network Security Administrator role has been defined, taking explicit ownership of continuous firewall and IDS/IPS tuning.

Vendor Risk Management: Review service level agreements (SLAs) with the current Internet Service Provider (ISP) to ensure upstream DDoS mitigation capabilities are enabled for future high-volume attacks.

Identify

NIST CSF — ID

Attack type:

ICMP Flood
Denial-of-Service (DoS)

Attack Vector:

External network —
spoofed or direct ICMP echo requests

Systems & assets affected

Internal network infrastructure
All internal network services

Perimeter Firewalls
Staff Workstations

Root cause

The perimeter firewall was improperly configured and lacked rules to rate-limit or block incoming ICMP packets from external sources. This allowed the attacker to send a high-volume flood of ICMP pings directly into the internal network without restriction.

Estimated impact

All internal network services were unavailable for approximately 2 hours. Business operations relying on internal systems — including client web design, graphic design, and social media workflows — were suspended during this period. No data exfiltration has been identified at this time.

Protect

NIST CSF — PR

Immediate controls implemented :

ICMP rate-limiting firewall rule
Source IP verification (anti-spoofing)
IDS/IPS deployment

Recommended additional protections

1. Conduct a full firewall configuration audit on a quarterly basis to identify any misconfigured or missing rules before they can be exploited.
2. Implement network segmentation so that a flood targeting one segment cannot propagate across the entire internal network, limiting blast radius in future incidents.
3. Establish a formal access control policy restricting which internal devices are permitted to send or receive ICMP traffic.
4. Deliver cybersecurity awareness training to all staff, covering how to recognize network disruptions and the proper internal reporting procedure.

Detect

NIST CSF — DE

Detection methods in place

Following the incident, the organization deployed network monitoring software to continuously track traffic patterns and flag anomalies. The IDS/IPS system now filters suspicious ICMP traffic based on behavioral signatures and volume thresholds.

Recommended detection improvements

1. Integrate a SIEM (Security Information and Event Management) platform to centralize log collection from the firewall, IDS/IPS, and network devices, enabling faster correlation of threat signals.
2. Configure automated alerts for abnormal ICMP packet volume exceeding baseline thresholds, triggering notification to the security team in real time.
3. Maintain an authorized device and user inventory, enabling rapid detection of unauthorized devices or accounts on the network.
4. Schedule regular reviews of firewall and IDS/IPS logs to identify patterns that automated systems may miss.

Respond

NIST CSF — RS

Incident response actions taken

Upon detection, the incident management team immediately blocked all incoming ICMP packets at the firewall, halted all non-critical network services to reduce load on the network, and focused resources on restoring critical services. The team isolated the perimeter firewall for investigation and documented all traffic logs from the attack window.

Response plan for future incidents

1. Containment: Upon detecting abnormal ICMP or other flood traffic, the on-call analyst will immediately apply a block rule at the firewall level and isolate affected network segments to prevent lateral spread.
2. Notification: The incident response lead will notify senior management and relevant stakeholders within 30 minutes of confirmed incident classification. External parties (clients, vendors) will be informed if service disruption is expected to exceed one hour.
3. Evidence collection: All firewall logs, IDS/IPS alerts, and traffic captures from the incident window will be preserved for forensic analysis and potential legal action.
4. Post-incident review: Within 5 business days of resolution, the cybersecurity team will conduct a formal lessons-learned meeting and update the incident response runbook accordingly.

Recover

NIST CSF—RC

Recovery actions taken

Critical network services were restored as the highest priority once ICMP traffic was blocked. Non-critical services were brought back online incrementally and verified for normal operation before being returned to full availability. Network traffic was monitored closely for 24 hours post-recovery to confirm the attack had been fully contained.

Recovery priorities (future incidents)

1. Restore internet-facing and client-critical services first (web hosting, client portals).
2. Restore internal collaboration and communication systems to enable coordinated response.
3. Verify the integrity of all restored services before resuming full operations.
4. Conduct a data integrity check to confirm no data loss or corruption occurred during the disruption period.

Process improvements

Establish formal recovery time objectives (RTOs) for each service tier, so the team has a defined target for how quickly each system must be restored. Maintain regularly tested backups of all critical configurations, including firewall rules, so they can be redeployed rapidly after an incident.